

# SYNOPSIS

## Razorpay AI Buildathon — Open Track

**Project Title:** Merchant Decision Intelligence Agent

**Working Name:** MerchantIQ — AI-Powered Merchant Decision Intelligence & Action System

### 1. Executive Summary

MerchantIQ is an AI-powered decision-intelligence system designed to help merchants understand what is happening in their payment-driven business, identify the reasons behind important changes, predict emerging risks and opportunities, and recommend the next best action. Unlike a conventional dashboard or chatbot, the system combines transaction analytics, machine learning, anomaly detection, forecasting, explainable insights and an agentic decision layer into one workflow: **Observe** → **Diagnose** → **Predict** → **Recommend** → **Approve** → **Act** → **Measure**.

### 2. Problem Statement

Small and growing merchants generate large volumes of payment and customer data but often lack the analytical capability to continuously interpret it. A dashboard may show that revenue has declined, but it does not necessarily explain the underlying drivers, quantify the business impact, predict what is likely to happen next, or prioritize the most valuable intervention. The proposed solution addresses this gap by converting raw payment and merchant data into evidence-backed decisions.

### 3. Proposed Solution

- Automated merchant KPI monitoring across revenue, transactions, average transaction value, payment success, refunds and customer activity.
- Anomaly detection to identify unusual changes in transaction volume, payment failures, refunds or merchant performance.
- Machine-learning models for short-term revenue forecasting and selected risk/opportunity predictions.
- Root-cause analysis that connects changes in business KPIs with contributing transaction, payment-method, customer or product patterns.
- An LLM-powered reasoning layer that converts analytical outputs into concise, evidence-based explanations.
- A decision agent that ranks recommended actions according to estimated impact, confidence and business priority.
- A human-approval and audit layer before any consequential action is executed.
- Outcome measurement so that recommendations can be evaluated against subsequent business results.

### 4. Objectives

- Build an end-to-end merchant analytics and decision-intelligence prototype.
- Apply Data Science techniques including EDA, statistical analysis, feature engineering and KPI analysis.
- Apply ML for anomaly detection and forecasting, with measurable validation metrics.
- Use GenAI/LLM reasoning to explain model and analytics outputs using supporting evidence.
- Generate prioritized, actionable recommendations rather than descriptive dashboards alone.

- Demonstrate safe agentic behavior through bounded actions, human approval and an audit trail.
- Quantify business value through metrics such as forecast error, anomaly-detection quality, estimated revenue at risk/recovered, and recommendation outcomes.

## 5. Proposed System Architecture

| Layer      | Components                                                                               |
|------------|------------------------------------------------------------------------------------------|
| Data       | Razorpay test/synthetic payment data, transaction history, merchant/customer attributes  |
| Analytics  | SQL/Python pipelines, EDA, KPI computation, cohort/trend analysis                        |
| ML         | Anomaly detection, revenue forecasting, optional churn/risk model                        |
| Reasoning  | LLM/RAG layer grounded in computed metrics, model outputs and transaction evidence       |
| Agent      | Diagnosis, prioritization, recommendation and approval workflow                          |
| Action     | Bounded test-mode/API actions where supported; otherwise simulated action with audit log |
| Interface  | Streamlit dashboard + natural-language query interface                                   |
| Evaluation | ML metrics, business-impact metrics, latency/throughput and failure-case testing         |

## 6. Data & Features

**Transaction-level:** transaction ID, timestamp, amount, payment method, status, refund status, merchant ID, customer ID and relevant test-mode attributes. **Derived features:** rolling transaction volume, success/failure rate, average transaction value, refund ratio, customer recency/frequency, merchant-level trends, time-of-day patterns and payment-method performance. The prototype will use Razorpay-provided test-mode/API data where available and clearly labeled synthetic or public data for fields that are not available.

## 7. Machine Learning Methodology

- **Anomaly detection:** compare Isolation Forest and statistical/rolling-baseline approaches; evaluate precision/recall where labeled anomalies are available and review false-positive rates.
- **Forecasting:** establish a naive/seasonal baseline and compare it with an appropriate time-series or ML forecasting model using MAE, RMSE and MAPE where suitable.
- **Optional customer model:** predict likelihood of reduced activity using a classification model, evaluated with precision, recall, F1 and calibration.
- **Explainability:** use feature importance/SHAP-style explanations where appropriate and pass structured evidence to the LLM rather than allowing unsupported claims.

## 8. Analytics & AI Layer

The analytics layer will produce a merchant health score, KPI trends, anomaly alerts and estimated business impact. The LLM will not replace the analytical models; it will interpret their structured outputs. A retrieval/grounding layer can provide definitions, merchant-specific context and model evidence. The agent will generate recommendations only from available evidence and will explicitly indicate uncertainty or insufficient data.

## 9. Example User Journey

**Scenario:** A merchant asks, “Why did my revenue fall this week and what should I do?”

**Step 1:** Analytics detects an 18% week-over-week decline.

**Step 2:** ML/anomaly modules identify increased payment failures and reduced repeat-customer activity.

**Step 3:** The system estimates the contribution of each driver and forecasts near-term revenue.

**Step 4:** The AI explains the evidence and ranks actions by expected impact.

**Step 5:** The merchant approves a bounded action in test mode/simulation.

**Step 6:** The system records the decision and measures the resulting outcome.

## 10. Evaluation Plan

| Area              | Primary metrics                                                    |
|-------------------|--------------------------------------------------------------------|
| Forecasting       | MAE, RMSE, MAPE/appropriate scaled error                           |
| Anomaly detection | Precision, recall, F1, false-positive rate                         |
| Analytics         | KPI correctness, reconciliation of computed totals, query accuracy |
| LLM reasoning     | Groundedness, evidence coverage, unsupported-claim rate            |
| Recommendations   | Acceptance rate, estimated impact, outcome improvement             |
| System            | Latency, throughput, failure handling, audit completeness          |

## 11. Expected Novelty

The novelty is not a generic “chat with your data” interface. The proposed system links quantitative analytics and ML predictions to an evidence-grounded decision agent. Its distinctive workflow is designed to answer four questions together: **What changed? Why did it change? What is likely to happen next? What action has the highest expected value?** The addition of approval, auditability and outcome measurement makes the system a decision loop rather than a static analytics tool.

## 12. Technology Stack

**Python, Pandas, NumPy, scikit-learn, SQL/MySQL, Streamlit, Plotly/Matplotlib, Git/GitHub, Razorpay APIs/test mode where applicable, and an LLM/RAG framework as required.** The final stack will be kept intentionally lightweight so that the prototype remains reliable and demonstrable.

## 13. Deliverables

- Working Streamlit prototype.
- GitHub repository with clean source code and README.
- Dataset/data-generation and preprocessing pipeline.
- Trained ML models with evaluation results.
- Merchant analytics dashboard.
- AI decision/recommendation interface.
- Architecture diagram and technical documentation.
- Test cases, failure scenarios and audit log.
- 5-minute buildathon demonstration/pitch.

## 14. Scope Control

To avoid an over-broad Open Track submission, the MVP will focus on three core capabilities: **(1) merchant performance diagnosis, (2) revenue/anomaly prediction, and (3) evidence-backed next-best-action recommendations.** Additional actions and models will be treated as extensions only after the core loop is stable and measurable.

## 15. Expected Outcome

The final prototype is expected to demonstrate that payment data can be transformed into timely, explainable and measurable business decisions. Success will be demonstrated through working analytics, validated ML performance, grounded AI responses, safe action handling and quantified business impact rather than through conversational capability alone.

**Note:** This synopsis is a proposed project design for the Razorpay Open Track. Exact Razorpay API availability, permitted actions, submission requirements and evaluation criteria should be checked against the current official buildathon instructions before final submission.